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Week: 6

Web and Mobile Application Development

1.

In a university environment, students and staff often experience technical problems such as Wi-Fi connection issues, password resets, and software installation errors. To improve support services, the IT department decides to develop a mobile application that allows users to report problems directly from their smartphones. The application is designed to be simple and easy to use.

Users log in using their student credentials. After logging in, they can create a support ticket by selecting a problem category, describing the issue, and uploading screenshots if needed. The request is then sent to the IT support team. Technicians can view the ticket, update the progress, and communicate with the user.

The app also sends notifications to inform users when their issue is received, being processed, or resolved. This system improves communication, reduces waiting time, and keeps records of all support requests. The mobile app makes IT support more organized and efficient.

2.

React Native is a framework used to build mobile applications using JavaScript and React. It allows developers to create applications for both Android and iOS using a single codebase. This means developers do not need to write separate code for each platform.

React Native uses native components, so the application looks and feels like a normal mobile app. It also supports hot reloading, which allows developers to see changes instantly without restarting the app.

React Native reduces development time and cost because one team can build applications for multiple platforms. It is popular in modern development because it is efficient, flexible, and supported by a strong community.

3.

To practice Flutter, I first downloaded and installed the Flutter SDK from the official website. After installation, I added Flutter to the system environment path and ran the flutter doctor command to check if all tools were installed correctly.

I then created a new project using the command flutter create sample_app. After opening the project in a code editor, I ran the application on an Android emulator. The default sample app displayed a counter button that increases a number when pressed.

From this activity, I learned that Flutter provides ready-made widgets that make user interface design easier. The setup process requires careful installation steps, but once completed, building and testing applications becomes simple and efficient.

4.

In Draw.io, the mobile app process flowchart begins with the user opening the application. The next step is authentication, where the user logs in using a username and password.

After successful login, the user selects an action such as creating a support ticket. The user enters the details and submits the request. The request is sent to the server and stored in a database.

The IT support team reviews the request and updates the ticket status. A notification is sent back to the user to show progress. The process ends when the issue is resolved. This flowchart clearly shows the step-by-step process of how a mobile application handles user requests.